

Momentum vs. Mean Reversion, Across Regimes, After Costs

Martingale · Research Note 003 Author: Neil Gilani · Reproducible code: [experiment.py](#)

Abstract

We compare two opposing trading ideas — **momentum** (bet the trend continues) and **mean reversion** (bet the last move reverses) — on 4,916 trading days of SPY (2007–2026), splitting the record into bull and bear regimes and charging realistic costs. After 1 bp costs, **neither beats buy-and-hold** (Sharpe 0.63): momentum nets 0.15, mean reversion 0.31. But the regime split is decisive — momentum leads in **bull** markets (0.20 vs. −0.25) while mean reversion dominates in **bear** markets (**1.07** vs. 0.17). And because mean reversion trades about **six times as often** (turnover 1.01 vs. 0.18), costs erase its edge fast: its Sharpe falls from 0.44 gross to **−0.21** at 5 bps. The lesson: neither idea is universally right — each is a conditional bet on a regime, and turnover decides which one survives costs.

1. Introduction

Two of the oldest instincts in markets flatly contradict each other. **Momentum** says a rising asset tends to keep rising — buy strength, sell weakness. **Mean reversion** says moves overshoot and snap back — buy weakness, sell strength. Both have made money; both have blown up. So the interesting question isn't "which is right?" but "**when is each right, and does the edge survive the cost of trading it?**"

This note tests both on SPY, with three disciplines that separate a real result from a backtest fantasy (all part of this lab's [methodology](#)):

1. **Regimes.** Markets behave differently when trending up vs. falling. We label each day *bull* (price at or above its 200-day average) or *bear* (below) and measure each strategy *within* each regime, not just on average.
2. **Costs.** A signal that only works for free isn't a strategy. We charge a realistic per-trade cost and report results at 0, 1, and 5 basis points.
3. **No lookahead.** Every position is decided from data up to day t and earns day $t+1$'s return — the exact error dissected in [Note 001](#).

Hypothesis (stated before running it): momentum works better in trending *bull* regimes, mean reversion better in *bear* / choppy regimes; and mean reversion — which flips position far more often — loses much more of its gross edge to costs. A result matching this supports the "each idea is a bet on a regime" view; a strategy that wins everywhere after costs would refute it.

2. Data & Method

Data. Daily closing prices for SPY (an ETF tracking the S&P 500) from January 2007 to July 2026 — 4,916 trading days, chosen to include the 2008 crash so that both a violent bear market and long bull runs are in the sample — downloaded via `yfinance`.

Signals. Each day t produces a position (+1 long or -1 short) held into day $t+1$, using only information available through day t :

- **Momentum:** +1 if the trailing 20-day return is positive, else -1.
- **Mean reversion:** +1 if yesterday's return was negative, else -1.

Regime. Day t is *bull* if that day's closing price is at or above its trailing 200-day average, else *bear*.

Costs. Each day we charge `cost × |change in position|`. A full flip from -1 to +1 moves 2 units, so at 1 bp per unit it costs 2 bps. We also report the average daily **turnover** of each strategy, which explains its sensitivity to costs.

Metric. Annualized Sharpe ratio — average daily strategy return ÷ its standard deviation, $\times \sqrt{252}$. Reported gross and net of costs, overall and within each regime, with **buy & hold** as the benchmark.

3. Results

Overall (1 bp costs):

Strategy	Gross Sharpe	Net Sharpe	Avg turnover
Buy & hold	0.625	0.625	0.000
Momentum	0.171	0.148	0.178
Mean reversion	0.437	0.307	1.010

Neither timing strategy beats simply buying and holding the index. Mean reversion (net 0.31) edges out momentum (net 0.15), but both trail buy & hold's 0.63.

After-cost Sharpe by regime:

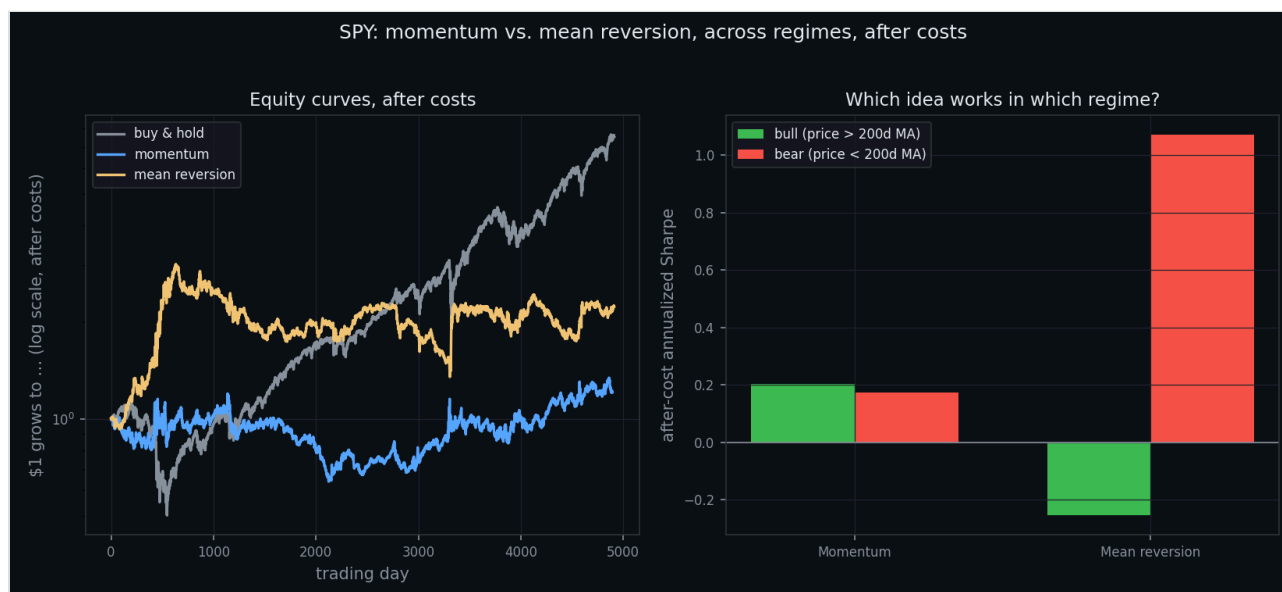
Regime	Momentum	Mean reversion
Bull (price $\geq$ 200d MA)	+0.204	-0.254
Bear (price $<$ 200d MA)	+0.173	+1.072

This is where the two ideas separate. In bull markets momentum is positive and mean reversion loses money. In bear markets the order reverses hard: mean reversion earns an after-cost Sharpe of **1.07**, more than six times momentum's.

Cost sensitivity (net Sharpe):

Cost	Momentum	Mean reversion
0 bps	0.171	0.437
1 bp	0.148	0.307
5 bps	0.057	-0.212

Mean reversion looks best when trading is free, but its edge collapses as costs rise — turning *negative* by 5 bps. Momentum, which trades far less, fades gently.



4. Discussion

Three findings, in order of how much they matter.

1. Neither beats buy-and-hold after costs. The single most useful number here is the benchmark: holding SPY netted a Sharpe of 0.63, and neither timing strategy came close (momentum 0.15, mean reversion 0.31). A daily long/short signal on one index, traded against the market's strong upward drift, is fighting a stiff headwind. This is the honest headline, and it is the result most "strategy" pitches quietly omit.

2. But each idea is a bet on a regime. The overall averages hide a clean split. Momentum is the better of the two in **bull** markets (+0.20 vs. -0.25); mean reversion is dramatically better in **bear** markets (+1.07 vs. +0.17). This matches the hypothesis and makes economic sense: bear markets are volatile and choppy — sharp drops followed by sharp bounces — which is exactly the environment short-term reversal is built for, while calm uptrends reward simply staying long with the trend. Mean reversion's strong overall net Sharpe (0.31) is really a bear- market phenomenon leaking through; in bull markets, which dominate the calendar, it actively loses money.

3. Turnover decides who survives costs. Mean reversion flips position almost every day (turnover 1.01) while momentum rarely changes its mind (0.18) — roughly a **6×** difference. That single fact drives the cost table: from 0 to 5 bps, mean reversion's Sharpe craters from 0.44 to −0.21 (a swing of 0.65), while momentum only slips from 0.17 to 0.06. A high-turnover edge is a mirage unless you can trade extremely cheaply; the same signal is a "strategy" for a low-cost market maker and a slow bleed for a retail account. This is Note 001's warning in a new form: an in-sample number that ignores costs is not a result.

Putting it together. There is no universal winner. Momentum and mean reversion are *conditional* bets — each earns its keep in one regime and gives it back in the other — and once realistic costs are charged, the higher-turnover idea keeps far less of its paper edge. The practical reading is not "trade mean reversion in bear markets" (regimes are only known with certainty in hindsight, and 5 bps is optimistic for a retail trader); it is that **strategy performance is inseparable from regime and from cost**, and any backtest that reports one number without both is telling you very little.

5. Limitations

This study is deliberately simple. It uses one instrument (SPY), one momentum lookback (20 days), one reversal horizon (1 day), and one regime definition (the 200-day average); other choices could give different numbers, and testing many and reporting the best would be the data-snooping error of Note 001 — so these were fixed in advance. The cost model is a flat per-turnover charge and ignores slippage, borrowing costs for shorts, and the fact that shorting an index isn't free — so the real-world net Sharpes are, if anything, *worse* than shown. The regime label uses the 200-day average known on the same day, which is realistic, but acting on it still requires knowing you are in a bear market, which is only obvious afterward. Sharpe says nothing about crash risk or the depth of drawdowns. A positive after-cost Sharpe here would be evidence a signal exists, not proof it is tradable at size. And past regime behavior need not repeat.

6. Conclusion

We asked whether momentum or mean reversion works on SPY, whether the answer depends on the market regime, and whether either survives realistic costs. Across 4,916 days, **neither beat buy-and-hold after costs** — but the two ideas split cleanly by regime: momentum led in bull markets, while mean reversion earned a strong after-cost Sharpe of 1.07 in bear markets before its high turnover let costs eat it alive at higher fee levels. Neither momentum nor mean reversion is universally right. Each is a bet on a regime, and once you charge for trading, the strategy that trades the most keeps the least.

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How to cite

Gilani, N. (2026). *Momentum vs. Mean Reversion, Across Regimes, After Costs*.
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